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# Lab 06: Jenkins Live

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# Objective

Run a real controller + agent with Docker Compose, connect them, and execute the canonical three-stage Jenkinsfile.

## Prerequisites

- Docker; concepts: Jenkins Setup, Jenkins Pipelines.

## Architecture

Compose: jenkins (controller, port 8080, volume jenkins\_home) + agent (inbound, label docker-agent) → Multibranch or Pipeline job → Build/Test/Deploy stages.

## Setup

```
mkdir jenkins-lab && cd jenkins-lab # docker-compose.yml: jenkins/jenkins:its + inbound agent joined with the secret from the controller UI
docker compose up -d
```

Get the admin password: `docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword`.

## Step 1 — Connect the agent

### Execute

Manage Jenkins → Nodes → create node agent-1, labels docker-agent, launch via inbound; run the agent container with the secret + controller URL. Install suggested plugins.

## Expected Result

Node page shows agent-1 online with 2–4 executors.

# Why

Controller schedules, agents execute — this split is the whole architecture (see Jenkins Architecture).

## Step 2 — Run the canonical pipeline

### Execute

New Pipeline job, agent docker-agent, script:

```
pipeline { agent { label 'docker-agent' } stages { stage('Build') { steps { sh 'echo build' } }  
stage('Test') { steps { sh 'echo test' } } stage('Deploy'){ steps { echo 'deploy' } } } }
```

## Expected Result

Stage View shows three green boxes on agent-1.

## Why

The minimal pipeline that proves scheduling, execution, and visualization all work.

## Verification

- [ ] Agent online, labeled, executors idle after run.
- [ ] Three green stages in Stage View.
- [ ] Controller ran zero builds (check: no executors on master).

## Failure Scenarios

- Agent offline: wrong secret/URL or controller unreachable — check container logs, use `http://jenkins:8080` in-Compose.
- No executors on controller is correct — never enable them in prod.

## Cleanup

`docker compose down` keeps `jenkins_home` volume; `down -v` wipes everything.

## What This Proved

Controller/agent split hands-on. Continue with Lab 07.

## Interview Notes

- `down` vs `down -v`: which one wipes `jenkins_home` and why that distinction is the whole lesson.

- Zero executors on the controller — what you check to prove no build ever ran there.

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